

# JYOTIRMAY SRIVASTAVA

✉ jyotirmaysrivastava.in@gmail.com · 📞 +91 9807865371 · 🌐 ·

📍 A4/3 Sulabh Awas Sector-1, Gomti Nagar Extension, Lucknow, Uttar Pradesh, India. Pincode:226010

## EDUCATION

**MS by Research - IISER Thiruvananthapuram** Kerala, India

Aug 2020 ▶ July 2024

3 year Program + 1 year Research

School of Biology

CGPA: 8.6/10

Research Focus: Neuroscience

Courses: Molecular Biology, Neuroscience, Animal Physiology, Structural Biology, Genetics, etc.

Research: Parkinson's Disease research in Dr. Poonam Thakur's Lab (January 2022 - June 2024)

**BSc (Hons) - Chaudhary Charan Singh University, Meerut** Uttar Pradesh, India

Aug 2017 ▶ July 2020

School of Life Sciences

Major in Microbiology | Percentage: 73

Courses: Molecular Biology, Virology, Mycology, Bacteriology, etc.

## PUBLICATIONS

Kachappilly, N., **Srivastava, J.**, Swain, B. P., & Thakur, P. (2022). **Interaction of alpha-synuclein with lipids.** In *Methods in Cell Biology* (Vol. 169, pp. 43-66). Academic Press.

## SKILLS

- IHC/ICC, Microscopy (Confocal)
- Mouse handling (maintenance, Intraperitoneal (IP) / Subcutaneous (SC) injections)
- Stereotactic Surgery, Brain dissection
- Cloning, PCR
- Protein purification (SEC, Ion Exchange, Ni-NTA)
- Mammalian cell culture, Western blotting
- Programming (Python, R, Bash)

## AWARDS AND HONORS

|                                                                                                                        |      |
|------------------------------------------------------------------------------------------------------------------------|------|
| <b>Parkinson's Foundation</b> Visiting Research Fellowship                                                             | 2024 |
| <b>Department of Biotechnology</b> -Junior Research Fellowship                                                         | 2023 |
| Indian Institute of Technology (IIT) <b>Joint Admission Test for Masters (JAM)</b> : 57 <sup>th</sup> Rank (All-India) | 2020 |
| <b>CSIR National Eligibility Test</b> for Junior Research Fellowship (NET) : 99.6 Percentile Rank (All-India)          | 2020 |
| <b>TIFR Nationwide Entrance Examination for Biology (JGEEBILS)</b>                                                     | 2020 |

## PROJECTS AND INTERSHIPS

**Master's Thesis: Evaluation of the neuroprotective potential of 6-BIO treatment in a chronic mouse model of PD**

April 2022 ▶ April 2024

Evaluating a potential Parkinson's Disease treatment avenue

- Developing a more physiologically relevant mouse model of Parkinson's Disease
- Performing stereotactic injections in mouse brain to target substantia nigra, regular follow-ups for behavioural assessment to estimate neurodegeneration
- Studying the changes in key molecular players involved in PD pathogenesis and mechanism of action of the drug

**Skills Gained:** Mouse handling, survival surgeries, immunohistochemistry, confocal microscopy, western blotting

**Cross-species analysis of publicly available single-cell transcriptomics datasets**

July 2023 ▶ June 2024

Drawing insights from single-cell transcriptomics datasets and subsequent modelling of affected biochemical pathways to develop a better understanding of Parkinson's Disease pathology

- Identifying publicly available single-cell transcriptomics datasets focusing on the mid-brain region in both normal and PD-affected states
- Integration of the identified cross-species datasets
- Identifying differentially expressed features, as well other transcriptional level changes from the integrated dataset
- Modelling the neural biochemical pathways based on insights from the analysis of the transcriptional data

**Skills Gained:** Single-cell data analysis, R, bash

### **Building a GUI-enable deep learning based approach towards automating the rodent cylinder test**

February 2023 ▶ May 2024

- Designed a video recording setup to capture mouse movement and paw usage
- Feature engineering to better capture the spatial and temporal components of mouse movements and actions
- Evaluating different DL model architectures to best integrate these spatiotemporal features and fine-tuning models to achieve high accuracy
- Integrated the training to analysis pipeline into an easy to use graphical user interface (GUI)

**Skills Gained:** bash, machine/deep learning, GUI design

### **Remote Internship: Dr. Sukant Khurana at CSIR-CDRI**

Jul 2019 ▶ Dec 2019

Learnt bioinformatics and RNAseq data analysis

- Got introduced to the pathology of cancer and the role of various ion channels, especially HCN ion channels, in the progression of the disease
- Learnt RNA-seq data analysis, using live and novel RNA-seq datasets generated from wet-lab experiments in the lab
- Learnt the intricacies of taking raw-data and extracting useful insights from the same to guide further wet-lab experiments

**Skills Gained:** RNA-seq data analysis, Python, R

### **Summer internship: Dr. Sanjay at Credora Life Sciences**

Jul 2019 ▶ Oct 2019

Learnt mammalian cell culture and q-PCR

- Got introduced to mammalian cell culture, and basic cell-based techniques (immunocytochemistry, and MTT cell-viability assay)
- Learnt basic techniques in molecular biology (DNA/RNA isolation, PCR, and q-PCR)
- Designed and executed a short project to identify and check the efficacy of a new anti-neoplastic drug combinations

**Skills Gained:** Cell culture, molecular biology techniques

### **CERTIFICATES**

|                                            |                                      |      |
|--------------------------------------------|--------------------------------------|------|
| <b>NeuroAI</b>                             | Neuromatch Academy                   | 2024 |
| <b>Computational Neuroscience</b>          | Neuromatch Academy                   | 2022 |
| <b>Genomic Data Science Specialization</b> | John's Hopkins University (COURSERA) | 2021 |
| <b>Bioinformatics I&amp;II</b>             | Toronto University (COURSERA)        | 2019 |
| <b>Medical Neuroscience</b>                | Duke University (COURSERA)           | 2019 |

### **MEMBERSHIPS**

|                                       |                  |      |
|---------------------------------------|------------------|------|
| <b>Indian Academy of Neuroscience</b> | Permanent Member | 2022 |
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